EAS Station

Professional Emergency Alert System Platform
Software-Defined Drop-In Replacement for Commercial EAS Hardware

Transform Emergency Alerting

EAS Station is a cutting-edge, software-defined Emergency Alert System that replaces $5,000-$7,000 commercial hardware with powerful, flexible software running on commodity hardware like Raspberry Pi. Monitor, broadcast, and verify NOAA and IPAWS alerts with FCC-compliant SAME encoding, multi-source aggregation, and spatial intelligence.

Made for Broadcasters • Amateur Radio • Emergency Management

Powerful Features for Professional Alerting

Everything you need for reliable, compliant emergency alert processing

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🌐 Multi-Source Ingestion

Automatically aggregate alerts from NOAA Weather Service, FEMA IPAWS, and custom CAP feeds. Never miss a critical alert.

📻 FCC-Compliant SAME

Generate valid Specific Area Message Encoding (SAME) headers per FCC Part 11 regulations. Full EAS protocol support.

🗺️ Geographic Intelligence

PostGIS-powered spatial filtering with county, state, and custom polygon support. Target alerts to specific areas.

📡 SDR Verification

Automated broadcast verification using RTL-SDR or Airspy receivers. Confirm your alerts are transmitting correctly.

🔒 Built-in HTTPS

Automatic SSL/TLS with Let's Encrypt integration. Secure nginx reverse proxy included for production deployment.

🎨 Modern Web UI

Responsive Bootstrap 5 interface with real-time updates. Manage your entire system from any device.

Built for Real-World Use Cases

🎙️ Broadcasters

- ✓ Replace $5,000-$7,000 commercial encoders
- ✓ Multi-station coordination
- ✓ Automated compliance logging
- ✓ Integration with existing broadcast chains

📡 Amateur Radio

- ✓ Emergency communications testing
- ✓ Alert relay networks
- ✓ Training and education
- ✓ Field deployment exercises

🚨 Alert Distribution

- ✓ Custom alert distribution systems
- ✓ Geographic targeting and filtering
- ✓ Integration with existing infrastructure
- ✓ Multi-format alert delivery

💻 Developers

- ✓ CAP protocol experimentation
- ✓ Alert system research
- ✓ Custom integrations and plugins
- ✓ Open-source contribution

System Requirements

Compute:	Raspberry Pi 5 (8GB) or x86	**Software:**	Docker 24+, Python 3.11+
Database:	PostgreSQL 14+ with PostGIS 3+	**Memory:**	4GB RAM (8GB recommended)
Audio:	USB sound card or Pi HAT	**SDR (Optional):**	RTL-SDR v3 or Airspy